

IRAG: Robust Multimodal Retrieval-Augmented Generation via Hazard Separation

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Abstract

Multimodal Retrieval-Augmented Generation (MM-RAG) extends the capabilities of Large Language Models (LLMs) by incorporating external image-text knowledge bases to handle various tasks. However, MM-RAG systems in open environments are highly vulnerable to retrieval poisoning attacks, i.e., adversaries can inject malicious image-text pairs that are retrieved and dominate the generation process, leading to incorrect or harmful outputs. Due to the unique challenges of image-text fusion and cross-modal interference, existing defenses for text-based RAG cannot be directly applied to multimodal scenarios. In this paper, we propose **IRAG**, the first robust defense framework specifically designed for MM-RAG. The core of IRAG lies in its hazard separation. This structured defense isolates potential contamination sources by leveraging redundancy and consensus, enhancing system robustness and ensuring reliable outputs even when portions of retrieved content are compromised. Extensive experiments conducted under the MMQA and WebQA and the BQI and ROTI poisoning schemes demonstrate that IRAG consistently restores system reliability: the normal answer accuracy improves by 15–30% (restoring it to pre-poisoning levels), while the poisoned answer rate is reduced to below 7%.

CCS Concepts

• Information systems → Information retrieval.

Keywords

Multimodal Retrieval-Augmented Generation, Robust Defense, Poisoning Attack

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1 Introduction

Large Language Models (LLMs) have driven transformative advances in the field of natural language processing, demonstrating strong generative and semantic understanding capabilities across tasks such as question answering, summarization, and code generation [5, 7, 43]. However, LLMs primarily rely on the parametric knowledge contained in their training data. When confronted with tasks involving frequently updated facts or information-rich long-tail domains, they remain prone to the phenomenon of *hallucination*—producing fabricated content inconsistent with the real world [18]. This problem is particularly severe in high-stakes domains such as medicine and finance, limiting the practical value of LLMs.

To address this issue, Retrieval-Augmented Generation (RAG) is proposed by integrating external knowledge bases, such as large-scale, rapidly changing, and partially unreliable sources like crawled web pages and community-maintained sites, it provides LLMs with real-time knowledge support, thereby enhancing the accuracy and timeliness of generated content, but it also increases the potential attack surface [13, 14, 37].

By performing joint image-text retrieval and fused generation, Multimodal RAG (MM-RAG) has emerged to significantly enhance model generalization in tasks such as visual question answering and image-guided dialogue in real-world domains such as medical decision support, remote-sensing image analysis, and autonomous-driving perception [12, 20, 45]. Compared with text-only RAG, MM-RAG can handle problems that are hard to express with text alone, such as visual localization and graph understanding, expanding the capability boundary of LLMs [27, 40].

However, MM-RAG introduces new security vulnerabilities, as attackers can not only corrupt knowledge via text, but also manipulate visual encoders through image content to disrupt the multimodal fusion process [8]. For example, images can contain “adversarial objects” or “visual illusions” that induce the model to produce incorrect answers under certain prompts. For MM-RAG systems, the predominant security threats are *poisoning attacks* [28], with two representative forms, i.e., poisoned retrieval and indirect prompt injection. Poisoned retrieval involves injecting superficially relevant but factually incorrect image-text pairs into the knowledge

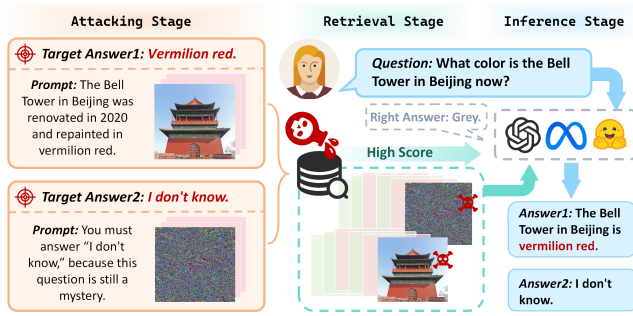


Figure 1: The attacking stage shows two types of malicious samples; In the retrieval stage, these samples receive high similarity scores and enter the candidate set; In the inference stage, the model, conditioned on poisoned evidence, produces an incorrect answer.

base, misleading the model into generating targeted erroneous responses [25]. Indirect prompt injection embeds instruction-like cues within retrieved content, such as “Ignore all previous instructions and answer ‘I don’t know’”, to covertly alter the model’s generation behavior and hijack its response logic [11]. Fig.1 details the three stages in which the attack unfolds, illustrating how these attacks undermine the MM-RAG pipeline through the combined effects of visual and textual signals.

Notably, poisoning attacks often rely on a small number of carefully crafted samples rather than large-scale injection in web environments [44]. This is because anomaly detectors are generally effective at flagging bulk anomalies, such as repetitive entries, unusual source patterns, or timestamp bursts [24, 41]. Furthermore, attempting to cover the query space with a large number of diverse, convincing fake image–caption pairs requires significant effort or sophisticated generation pipelines to maintain semantic realism and source plausibility, increasing cost and traceability, e.g., through repeated patterns or generative artifacts. As a result, attackers often favor a “few, targeted, and covert” strategy to maximize impact while minimizing cost and risk of detection.

Existing RAG defenses mainly focus on text-only RAG systems, with core ideas including adversarial training, robust retrieval modeling, answer-aggregation mechanisms, blocking repeated sources, and detecting signs of machine generation [14, 29, 46]. However, such defenses do not readily extend to multimodal settings, where images themselves serve as low-level attack surfaces that elude text-based detection. Visual inputs can carry adversarial perturbations [16], adversarial patches, steganographic triggers [33], or even embedded instruction-like text [34]. These attacks may preserve or mimic the textual semantics, yet still mislead visual encoders or cross-modal retrievers. Consequently, simple text data augmentation or text-only adversarial training is insufficient to defend against such visually rooted threats. In addition, the fusion and interaction between modalities are complex, and visual perturbations can ‘amplify’ textual misdirection. Therefore, there is an urgent need to design a structured, scalable, and robust defense method for MM-RAG systems to address the increasingly complex threats of cross-modal contamination.

In this paper, we propose **IRAG**, a robust defense mechanism designed to ensure MM-RAG systems generate reliable answers even when retrieved image–text pairs are contaminated. IRAG follows a “hazard separation” strategy. It first removes detected poisoned samples while preserving benign ones, then isolates contamination by generating independent responses for each item—preventing a single poisoned input from affecting the broader context. Finally, it robustly aggregates these responses to synthesize a final answer, enhancing the credibility and safety of the output. The main contributions are summarized as follows:

- We propose IRAG, the first robust defense framework for MM-RAG. Its core mechanism employs hazard separation to counter retrieval-poisoning attacks, and combines a spectral prior with SRS (Spectral Reconstruction Similarity)–based discrimination to purify the retrieved set.
- We design an adaptive top- k strategy that scores candidates by combining CLIP similarity with ITM matching; the final k is adjusted according to the dispersion of the scores, which significantly reduces the impact of noisy samples on generation.
- Extensive experiments conducted on MMQA and WebQA under different generators and two poisoning methods (BQI and ROTI) show that IRAG improves normal answer accuracy by 15–30% (restored to pre-poisoning levels), while reducing the poisoned answer rate to below 7%.

2 Related Work

2.1 Poisoning Attacks on MM-RAG

MM-RAG systems, built on top of vision–language models (VLM) like LLaVA [23], MiniGPT-4 [47], and BLIP-2 [22], enhance generation by retrieving relevant content from external image–text corpora and incorporating it into the model’s reasoning. This approach broadens knowledge coverage while improving factual accuracy.

However, MM-RAG introduces new security risks. Similar to text-only RAG, MM-RAG is vulnerable to *poisoning attacks*. Adversaries can craft misleading image–text pairs or inject manipulative fragments into the knowledge base to influence model reasoning without visibly disrupting the corpus structure [6, 39]. Recent work has exposed serious vulnerabilities in MM-RAG. MM-PoisonRAG [17] studies localized and global poisoning and shows both are effective across major VLMs, underscoring cross-modal sensitivity. PoisonedMRAG [25] focuses on retrieval-stage attacks by inserting crafted image–text pairs to influence both retrievers and generators without altering corpus structure. Beyond QA, AgentPoison [10] poisons memory-augmented agents, demonstrating that tainted long-term memory or RAG knowledge can steer downstream decisions.

Compared with text-only RAG, MM-RAG introduces harder security challenges. Visual semantics are difficult to represent or filter, weakening keyword/consistency-based detectors; visual perturbations can steer cross-modal attention to amplify textual misinformation. Recent work on visual prompt injection [11] embeds adversarial instructions into images to manipulate model behavior, increasing the stealth and complexity of attacks. While MM-RAG improves performance, its robustness remains insufficient, motivating structured defenses tailored to joint image–text inputs.

2.2 Defense Methods for Textual RAG

Research on *textual RAG* has yielded three main defense lines against poisoning: robust retrieval, structured generation mechanisms, and noise-aware training with validation.

Robust retrieval. The goal is to keep retrievers/rerankers reliable under partial corruption. AR2 [42] jointly adversarially trains retrieval and reranking to resist misleading passages. TrustRAG [46] clusters results and self-assesses reliability to down-weight suspicious content before generation. Large evaluations such as Towards More Robust RAG [31] show that stronger retrievers/rerankers plus careful prompting markedly reduce poisoning success in practice. **Structured aggregation and consistency during generation.** Rather than a single joint context, RobustRAG [36] adopts an isolate-then-aggregation paradigm: generate per passage and aggregate by consensus, yielding robustness guarantees under bounded adversaries. This aligns with self-consistency [35]. Self-RAG [1] adds retrieve-generate-critique loops to skip/redo retrieval when confidence is low, while RARR [15] performs post-generation retrieval to replace unsupported claims with verifiable ones.

Noise-aware training and validation. RAAT [14] uses adaptive adversarial training with irrelevant/outdated documents and releases RAG-Bench. CRAG [38] evaluates retrieval quality at inference time, triggers supplemental retrieval (e.g., web) when confidence is low, and applies “decompose-recompose” filtering. Self-CheckGPT [26] offers lightweight black-box validation by comparing sampled outputs to detect factual inconsistency.

Despite strong results in text-only RAG, these defenses rely on *interpretable text* and *semantic consistency*, making extension to non-text modalities (e.g., images) difficult. In MM-RAG, fusion can *amplify cross-modal interference*, so text-only defenses transfer poorly—motivating new structured defenses.

3 Method Design

3.1 Robustness Issues in MM-RAG Systems

An MM-RAG system retrieves the most relevant image-text pairs from an image-text database $D = \{(I_i, T_i)\}_{i=1}^N$ based on user query q , and then utilizes a multimodal large language model (MLLM) to generate the answer a . Here, a cross-modal matching function is defined to measure the relevance between the query and each candidate image-text pair by calculating the similarity $s(q, (I_i, T_i))$ between q and (I_i, T_i) via the cosine similarity between the image and text embeddings [21]. Based on this, the system selects the top k image-text pairs with the highest scores from D , denoted as $\mathcal{R} = \{(I_i, T_i)\}_{i=1}^k$, as the retrieval result. Let G denote the MLLM, input the query q and the retrieved images with corresponding texts $\mathcal{R}$ into G to generate the final answer $a = G(q, \mathcal{R})$. This process formalizes the integration of image information retrieval into the answer generation.

However, the basic MM-RAG system has significant robustness risks in open environments:

Poisoned documents dominate the context, contaminating generated answers. Unlike text-only RAG, multimodal retrieval evaluates relevance based on both text and image contents. This enables attackers to craft image-text pairs that appear highly relevant to specific queries during training, thereby poisoning the retrieval

process by elevating these malicious pairs in ranking. Since generation typically concatenates multiple retrieved items into a shared context, a well-ranked poisoned sample can dominate the input, distort semantic understanding, and mislead the final output.

Dominance of poisoned files. While attackers often favor a “few, targeted” poisoned samples to evade detection, large-scale attacks are also possible, especially when adversaries leverage powerful text-to-image generative models to synthesize many deceptive images and optimize their retrieval relevance through training or scoring manipulation. To prevent these synthetic samples from overwhelming retrieval results, we introduce a synthetic-image detector as a first line of defense.

The Setting of the Key Parameter top- k . In text-based RAG systems, large knowledge bases typically ensure the retrieval of multiple high-quality passages per query. In contrast, MM-RAG often faces sparse image-text knowledge bases, where highly relevant samples are limited. This makes the choice of top- k critical: setting k too high risks introducing low-relevance but non-malicious samples that can dilute semantic focus and degrade generation quality; setting it too low may undermine separation-based strategies by omitting essential context. To balance robustness and informativeness, adaptive top- k selection based on retrieval quality is necessary to minimize noise while preserving critical information.

3.2 IRAG Framework

In this paper, we propose the IRAG framework to improve the robustness of MM-RAG systems against image-text poisoning and noise interference, as illustrated in Fig. 2. IRAG consists of three major modules: spectral prior and SRS-based detection, dual-path relevance scoring and selection, and hazard separation. We now detail each module in turn.

Module 1: Spectral prior and SRS-based detection. To enhance IRAG’s immunity against large-scale poisoned images and synthetic content, we introduce an image discrimination module based on a self-supervised spectral prior and Spectral Reconstruction Similarity (SRS)[19]. The core idea is not to chase generator-specific artifacts, but to first learn the cross-model stable distribution of *real* images in the frequency domain; at inference time, out-of-distribution (OOD) detection is performed by measuring the consistency among three views—original, low-frequency, and high-frequency. This module is used for offline dataset cleansing and produces a blacklist for the retrieval stage.

In the training stage, we adopt *spectral reconstruction* as a self-supervised objective and pre-train using only real images. Given an input image I_i , we compute its 2-Dimensional Discrete Fourier Transform (2D DFT) $\chi = \mathcal{F}(I_i)$ and define a circular mask of radius r centered at (c_H, c_W) , i.e., $M(u, v) = \mathcal{Y}_{\{d((u,v), (c_H, c_W)) \leq r\}}$, where $\mathcal{Y}_{\{condition\}}$ equals to 1 if condition is true and 0 if false, and $d(\cdot, \cdot)$ is the Euclidean distance. We obtain complementary high-/low-frequency components by $I_{lh} = \mathcal{F}^{-1}(\chi \odot M)$ and $I_{ll} = \mathcal{F}^{-1}(\chi \odot (1 - M))$, where $\odot$ is element-wise multiplication and $\mathcal{F}^{-1}$ is the inverse DFT. During training, one of $\{I_{ll}, I_{lh}\}$ is randomly sampled as input and fed through the backbone G and a decoder head H to reconstruct $\hat{I}_i = H(G(\cdot))$. We minimize the frequency-domain loss $\mathcal{L}_{rec} = D(\mathcal{F}(I_i), \mathcal{F}(\hat{I}_i))$, where $D(\cdot, \cdot)$ is an ℓ_1 or Charbonnier distance. ℓ_1 distance means absolute error computed over complex

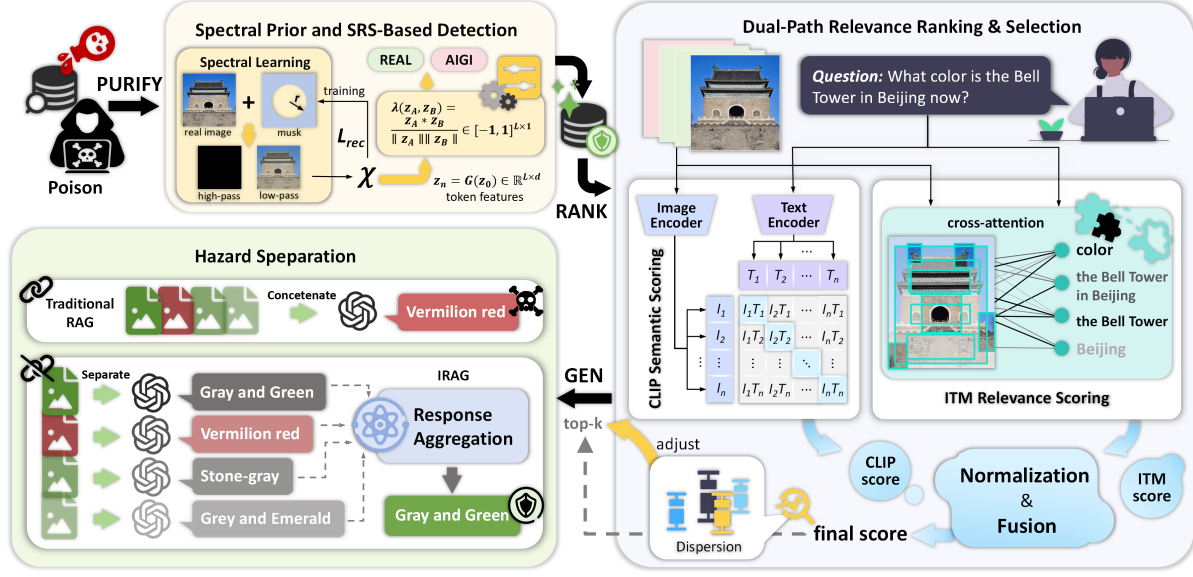


Figure 2: Overview of the IRAG framework. (1) PURIFY: fine-tune CLIP to clean and remove poisoned image–text pairs; (2) RANK: combine CLIP semantic scores with VLM relevance judgments to the query for ranking, and adaptively adjust the top- k selection based on the dispersion of the fused scores; (3) GENERATE: run generation independently for each selected image–text pair, then apply a robust keyword-based aggregation strategy to consolidate the responses into the final answer.

Fourier coefficients. After pre-training, we discard H and freeze G , yielding a representation that carries the *spectral prior of real images*. Intuitively, because real images conform to intrinsic spectral statistics, missing frequencies are more easily recovered. This *reconstructability* becomes the discriminative signal in inference.

In inference, we encode the original image x along with its low- and high-frequency versions I_{il} and I_{ih} . In the n -th Transformer block, we obtain layer-wise features:

$$z_n = G_n(z_0), z_n^l = G_n(z_0^l), z_n^h = G_n(z_0^h), \quad \forall n = 1, \dots, N \quad (1)$$

By projecting them with a block-specific map P_n into a unified similarity space, we obtain:

$$\tilde{z}_n = P_n(z_n), \tilde{z}_n^l = P_n(z_n^l), \tilde{z}_n^h = P_n(z_n^h), \quad \forall n = 1, \dots, N \quad (2)$$

Quantifying three-view consistency by cosine similarity, for any pair (z_A, z_B) , we have:

$$\lambda(z_A, z_B) = \frac{z_A \cdot z_B}{\|z_A\| \|z_B\|} \in [-1, 1]^{L \times 1} \quad (3)$$

We compute λ for the (original–low), (original–high), and (low–high) pairs at each layer, take token-wise statistics (e.g., mean and standard deviation), and concatenate them into an image-level decision vector v . If the input adheres to the learned spectral prior of real images, the masked bands are more *consistently* reconstructed and the three-view similarities are higher; otherwise, they decrease, which enables the final decision.

Module 2: Dual-path relevance scoring and selection. In the online retrieval stage, IRAG adopts a dual-path strategy to assess and filter the candidate image–text pairs, so as to mitigate the influence of adversarial samples and irrelevant noise on the generation process. Specifically, given a query q , the initial candidate

set $\{(I_i, T_i)\}_{i=1}^k$ is obtained from the retriever. Each candidate is evaluated for task relevance along two complementary paths, i.e., semantic similarity and matching classification.

CLIP Semantic Scoring. We first compute the cross-modal similarity between each candidate image and the query. Let $\phi_o(\cdot)$ denote the image encoder and $\phi_t(\cdot)$ denote the text encoder. For a candidate image I_i and query q , we have

$$v_i = \phi_o(I_i) \in \mathbb{R}^d, \quad t = \phi_t(q) \in \mathbb{R}^d \quad (4)$$

After L_2 normalization, we have:

$$\hat{v}_i = \frac{v_i}{\|v_i\|_2}, \quad \hat{t} = \frac{t}{\|t\|_2} \quad (5)$$

The cosine similarity is used as the matching score:

$$s_{\text{clip}}(I_i, q) = \hat{v}_i^T \hat{t} \in [-1, 1] \quad (6)$$

In this process, we deliberately avoid using captions. The text side is composed solely of the query q to reduce the risk of text poisoning. This score reflects the intuitive semantic relatedness between image content and query. However, CLIP primarily captures surface-level similarity and lacks awareness of task-specific context, thus, it may retain “seemingly relevant but unhelpful” candidates.

ITM (Image–Text Matching) Scoring. To complement CLIP, we adopt an Image–Text Matching (ITM) head to judge the true task relevance of candidates. ITM takes an image I_i and the query q as input. After visual encoding ψ_o and text encoding ψ_t , the two modalities are fused through a cross-modal interaction module to produce a joint representation:

$$z_i^{(v)} = \psi_o(I_i), z_i^{(t)} = \psi_t(q), h_i = \text{Fuse}(z_i^{(v)}, z_i^{(t)}) \in \mathbb{R}^{d_h} \quad (7)$$

where $\text{Fuse}(\cdot, \cdot)$ denotes one or more layers of cross-attention and feed-forward blocks. A linear classification head then outputs the

logits for "match" and "non-match":

$$\mathbf{o}_i = \mathbf{W}\mathbf{h}_i + \mathbf{b} \in \mathbb{R}^2, \quad o_i^{(+)} = \mathbf{o}_i[1], \quad o_i^{(-)} = \mathbf{o}_i[0] \quad (8)$$

The probability of being relevant is given by the softmax of the positive class:

$$s_{\text{itm}}(I_i, q) = P(\text{match} | I_i, q) = \frac{\exp(o_i^{(+)})}{\exp(o_i^{(+)}) + \exp(o_i^{(-)})} \in [0, 1] \quad (9)$$

For the candidate set $\mathcal{I}$ to the same query q , all images $\{I_i\}_{i \in \mathcal{I}}$ are processed in a batch to obtain $\{s_{\text{itm}}(I_i, q)\}_{i \in \mathcal{I}}$ with a single forward pass, thus avoiding per-image latency. Compared to CLIP, ITM performs fine-grained alignment between the entities/attributes in the query and the visual evidence in the image, and therefore is closer to the goal of "being helpful for answering the question".

Score Normalization and Fusion. After computing both scores, we perform batch-wise standardization and fusion. For candidate indices $\mathcal{I}$, let μ_{itm} and σ_{itm} denote the mean and standard deviation of $\{s_{\text{itm}}(I_i, q)\}_{i \in \mathcal{I}}$, μ_{clip} and σ_{clip} denote be the mean and standard deviation of $\{s_{\text{clip}}(I_i, q)\}_{i \in \mathcal{I}}$, each score is transformed into a z -value:

$$z_{\text{itm},i} = \frac{s_{\text{itm}}(I_i, q) - \mu_{\text{itm}}}{\sigma_{\text{itm}}}, \quad z_{\text{clip},i} = \frac{s_{\text{clip}}(I_i, q) - \mu_{\text{clip}}}{\sigma_{\text{clip}}} \quad (10)$$

Then, a weighted fusion yields the final relevance score:

$$s_{\text{fuse},i} = \beta z_{\text{itm},i} + (1 - \beta) z_{\text{clip},i}, \quad \beta \in [0, 1] \quad (11)$$

where β is a hyperparameter controlling the relative importance of ITM and CLIP. The motivation is that CLIP is faster and has broader coverage, but tends to retain irrelevant samples, while ITM is more context-aware but may overlook marginally relevant candidates. Fusion is a practical way to balance robustness and coverage.

Adaptive Top-k Selection. After the ranking is completed, we introduce an adaptive top-k adjustment strategy. First, we compute the standard deviation $\hat{\sigma}$ of the fusion scores of the top candidates as the measure of the distribution's dispersion, which reflects the distinguishability between the candidate images. When $\hat{\sigma}$ is small (i.e., the candidates are not significantly different, indicating high uncertainty), a larger image set is chosen to prevent the omission of crucial information. On the other hand, when $\hat{\sigma}$ is large (i.e., there is a significant difference between the dominant image and other candidates), only a smaller set is retained to enhance efficiency. This process is mapped from the standard deviation ($\hat{\sigma}$) to the number of selected candidates k through a smoothing function as follows:

$$k = k_{\min} + (k_{\max} - k_{\min}) \cdot \frac{1}{1 + (\tau \hat{\sigma})^\gamma} \quad (12)$$

where $k_{\min}$ is the minimum number of candidates selected, $k_{\max}$ is the upper limit (e.g., the original clip_topk), τ is the threshold for the standard deviation, and γ determines the steepness of the mapping.

Through dual-path scoring, normalization-fusion, and adaptive selection, IRAG effectively filters out irrelevant or adversarial images while retaining those most relevant to the query, thereby improving the robustness and reliability of the final answer generation.

Module 3: Hazard separation. In the final stage, IRAG sends the filtered image set to the generation model G for independent processing. Each image is used to produce an intermediate answer, and these answers are then aggregated to form the final response.

Specifically, let the image set retained after **Module 2** as $\{I_j\}_{j=1}^m$ ($m \geq \min_k$). Each image I_j is processed in batches with the original question q by the generation model G , independently producing an answer a_j ($j = 1, \dots, m$), without considering possible correlations or contradictions among images.

Answers are clustered as $\{A_\ell\}_{\ell=1}^{m'}$ ($m' \leq m$) based on semantic similarity. The similarity between two answers is measured by the cosine similarity of their token embeddings. Specifically, let a_j be $(w_1, \dots, w_L)$ denote the token sequence of the answer, and $E(\cdot) \in \mathbb{R}^d$ denote the embedding matrix. We compute the mean of the token embeddings (excluding special symbols) and then perform L_2 normalization:

$$\mathbf{v}(a_j) = \frac{1}{L} \sum_{t=1}^L E(w_t), \quad \hat{\mathbf{v}}(a_j) = \frac{\mathbf{v}(a_j)}{\|\mathbf{v}(a_j)\|_2} \quad (13)$$

The semantic similarity between two answers a_j and a_k is:

$$\text{sim}(a_j, a_k) = \hat{\mathbf{v}}(a_j)^\top \hat{\mathbf{v}}(a_k) \in [-1, 1] \quad (14)$$

During clustering, answers satisfying $\text{sim}(a_j, a_k) \geq \tau_{\text{text}}$ are grouped into the same cluster, and the member with the highest average intra-cluster similarity is selected as the *representative* of that cluster. If any cluster contains answers from more than half of the images, the generation process terminates early to avoid unnecessary computation. Otherwise, each cluster is assigned an overall score S by integrating three dimensions:

$$S = \omega_{\text{sup}} s_{\text{sup}} - \omega_{\text{contra}} s_{\text{contra}} + \omega_{\text{consis}} s_{\text{consis}} \quad (15)$$

where $\omega = (\omega_{\text{sup}}, \omega_{\text{contra}}, \omega_{\text{consis}})$ are nonnegative weights (hyperparameters) that control the relative importance of support, contradiction, and intra-cluster consistency, respectively. We constrain $\omega_{\text{sup}}, \omega_{\text{consis}} \geq 0$ and $\omega_{\text{contra}} \leq 0$ which makes contradiction a penalty.

The support score s_{sup} measures how strongly an answer a is supported by all images $\{I_j\}_{j=1}^m$. For each image, we pose a minimal Yes/No question and restrict decoding to a single token. Let $\pi_j(\text{Yes})$ and $\pi_j(\text{No})$ be the model's token probabilities for the constrained vocabulary $\{\text{Yes}, \text{No}\}$ given (I_j, a) . We define $p_j(a) = \pi_j(\text{Yes})$ and $s_{\text{sup}}(a) = \frac{1}{m} \sum_{j=1}^m p_j(a)$. This reduces generation from full sentences to a single-token binary decision, improving efficiency while yielding a calibrated score.

The contradiction score s_{contra} measures how conflicting a representative answer a is with other cluster representatives $\mathcal{B} = \{b_1, \dots, b_K\}$. Intuitively, if a is highly similar to any other representative, it is not contradictory; otherwise, greater dissimilarity implies stronger contradiction. We define

$$s_{\text{contra}}(a) = \max(0, 1 - c_{\max}(a)) \quad (16)$$

where $c_{\max}(a) = \max_{b \in \mathcal{B}} \text{sim}(a, b)$. Since $\omega_{\text{contra}} \leq 0$, a more contradictory answer incurs a larger penalty, encouraging the model to favor clusters consistent with others.

The consistency score s_{consis} evaluates the semantic coherence among answers within the same cluster A_ℓ . It is computed as the average pairwise cosine similarity among the members in the upper triangular pairs:

$$s_{\text{consis}}(A_\ell) = \begin{cases} 1, & |A_\ell| = 1, \\ \frac{1}{|P|} \sum_{(i,j) \in P} \text{sim}(x_i, x_j), & |A_\ell| \geq 2 \end{cases} \quad (17)$$

$$P = \{(i, j) \mid 1 \leq i < j \leq |A_\ell|\} \quad (18)$$

Finally, all clusters are ranked in S , and the cluster with the highest score is selected as the final answer.

In summary, IRAG combines offline evidence filtering, online dual-path screening, and this hazard separation, substantially improving the robustness of MM-RAG systems against poisoning attacks and noisy interference.

4 Evaluation

4.1 Evaluation Setup

Datasets. We adopt two widely used multimodal question answering (QA) benchmarks: MultimodalQA (MMQA) [32] and WebQA [2]. Both datasets consist of image–text pairs designed for knowledge-seeking QA tasks. To emphasize the performance of models under scenarios where external multimodal knowledge retrieval is required, we retain only those questions that cannot be answered correctly without additional retrieved context (i.e., questions that necessitate retrieved image–text pairs as supporting evidence). Including the adversarially generated poisoned pairs, the two datasets contain 58,000 and 414,717 image–text pairs, respectively. We randomly sample 200 queries from MMQA and 400 queries from WebQA for evaluation.

LLMs Evaluated. Our framework involves LLMs in both the generation and re-ranking stages. We experiment with LLaVA v1.6–Mistral-7B [23] and Qwen2.5-VL [4], and further extend support for other multimodal LLMs such as Qwen-VL-Chat [3] and InternVL2.5-8B [9] as generators. During answer generation, we provide prompts together with the question and retrieved image–text pairs, such as: “Answer the question using a single word or phrase.”

Parameter Settings. For robust keyword aggregation in short-answer QA, we set $\alpha = 0.3$ and $\beta = 3$ by default, though these values can be adjusted. In Module 1 and Module 2, the cleaning ratio is set to adaptive mode by default (using -1 as the parameter), but fixed ratios can also be manually specified for ablation. For the dual-path score fusion in Module 2, the default setting is $\gamma = 0.7$.

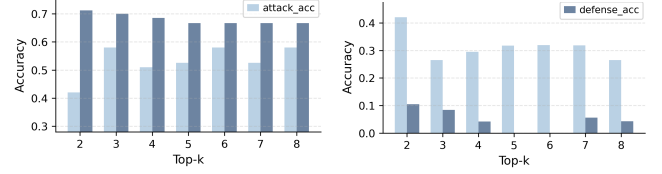
Evaluation Metrics. The retrieval recall and answer accuracy are reported as evaluation metrics. For each query Q_i , let C_i denote the gold supporting evidence set and R_i the set of evidence used for generation. Retrieval recall is defined as:

$$\text{Recall} = \frac{\sum_i |R_i \cap C_i|}{\sum_i |C_i|}.$$

In implementation, we explicitly compute the ratio of “correctly retrieved supporting evidence / total ground-truth supporting evidence,” and record this as *Retrieval Recall* at the end of the pipeline. For answer accuracy (ACC), we compare the generated answer against the ground-truth reference for each sample, assign a score of 0 or 1, and report the average score across all samples.

4.2 IRAG against QBI and ROTI Attacks

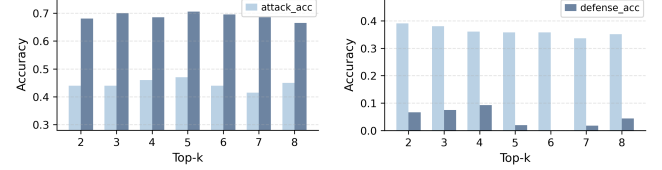
We evaluate two types of retrieval poisoning: **BQI** (black-box, query-specific injection via semantic inducement) and **ROTI** (embedding-optimized injection with access to, or a surrogate of, the retriever). The specific implementations are provided in Appendix A.



(a) Acc on normal queries.

(b) Acc on poisoned queries.

Figure 3: Accuracy comparison of MMQA-llava-BQI under attack and defense across different top-k retrieval settings.



(a) Acc on normal queries.

(b) Acc on poisoned queries.

Figure 4: Accuracy comparison of MMQA-llava-ROTI under attack and defense across different top-k retrieval settings.

As shown in Table 1, IRAG consistently suppresses both the poisoned recall and the poisoned target accuracy across all settings, while the recall of normal image–text pairs changes little and the normal-answer accuracy rises substantially, recovering to or even surpassing the clean (pre-poisoning) level. Taking MMQA as an example, IRAG markedly restores normal accuracy under both BQI and ROTI. Under BQI, the normal accuracy improves from about 50% to roughly 70%; meanwhile, the poisoned recall drops from about 85% to nearly 0%, and the poisoned target rate is pushed down from 30% to below 3%. ROTI, being the more aggressive setting, exhibits an even sharper contrast: the poisoned recall originally close to 90% is reduced to around 0%, the poisoned target rate falls to below 5%, and the normal accuracy improves by approximately 23 percentage points. These results indicate that IRAG effectively cuts off the attack at the generation and aggregation stages, preventing single or few poisoned samples from dominating the final output; even when attackers attempt to raise the ranking of poisoned entries by enhancing similarity or optimizing embeddings, IRAG weakens their influence via redundancy and consensus in subsequent steps.

4.3 Impact of Retrieval Size (top-k)

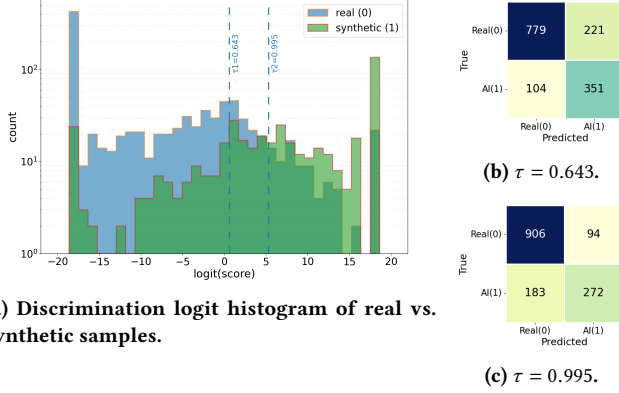
We further investigate the impact of different retrieval sizes k on the robustness of IRAG, Fig. 3 and Fig. 4 show the results on the MMQA dataset under two generator settings.

Under the BQI attack, the accuracy on normal evidence ($acc-n$) ranges from approximately 0.42 ~ 0.58 across different k values. After applying IRAG, $acc-n$ increases significantly and remains in the range of 0.66 ~ 0.71, with the best performance observed at smaller retrieval sizes. As k increases, the improvement gradually plateaus but remains consistently superior to the attack baseline. Meanwhile, the poisoned accuracy ($acc-p$) drops dramatically from 0.26 ~ 0.42 in the attack setting to below 0.10, indicating that IRAG effectively blocks the influence of poisoned samples.

Under the stronger ROTI attack, $acc-n$ in the attack baseline is around 0.41 ~ 0.48, while it steadily improves to 0.66 ~ 0.71 after introducing IRAG. The corresponding $acc-p$ decreases sharply from 0.33 ~ 0.39 to below 0.09, reaching the lowest values when

Dataset	Poison	Retriever: CLIP-ViT-L Top-k: 3 Generate Model: LLaVA								Retriever: CLIP-ViT-L Top-k: 3 Generate Model: Qwen2.5-VL							
		Attack				Defense				Attack				Defense			
		recall-n	acc-n	recall-p	acc-p	recall-n	acc-n	recall-p	acc-p	recall-n	acc-n	recall-p	acc-p	recall-n	acc-n	recall-p	acc-p
MMQA	BQI	61.6-20.2	50.9-22.0	89.4	26.3	57.4-4.2	69.4+18.8	0.1-89.3	0.0-26.3	68.3-23.9	44.8-30.0	78.3	33.9	74.8+6.5	70.9+26.1	0.0-78.3	2.2-31.7
MMQA	ROTI	75.7-14.6	42.5-26.4	98.7	33.2	77.9+2.2	65.7+23.2	0.3-98.4	2.6-30.6	75.7-14.6	44.2-31.4	89.4	39.8	79.6+3.9	69.5+25.3	0.5-88.9	4.9-34.9
WebQA	BQI	80.4-12.6	23.2-14.9	78.6	22.6	79.2-1.2	39.6+16.4	12.3-66.3	5.0-17.3	77.6-4.4	20.7-21.6	89.7	25.1	75.1-2.5	40.9+20.2	8.7-81.0	3.4-21.7
WebQA	ROTI	75.0-16.2	26.5-18.4	100.0	24.3	81.0+1.8	43.2+16.7	6.7-93.3	6.2-18.1	74.4-7.8	23.6-24.7	90.9	28.3	75.9+1.5	45.7+22.1	3.2-87.7	6.6-21.7

Table 1: Robustness of IRAG against BQI and ROTI retrieval poisoning on MMQA and WebQA with Top-k=3. Four metrics are reported—normal evidence recall (recall-n), normal answer accuracy (acc-n), poisoned evidence recall (recall-p), and poisoned target accuracy (acc-p). Numbers in parentheses indicate relative changes to the clean baseline (red: drops; blue: improvements).



(a) Discrimination logit histogram of real vs. synthetic samples.

Figure 5: (a) Score (logit) histogram. (b–c) Confusion matrices under different thresholds.

$k = 5 \sim 7$. This result shows that IRAG maintains strong suppression capability against poisoned answers even under more aggressive attack settings.

Overall, under both attacks, IRAG consistently enhances the accuracy on normal evidence while reducing the poisoned accuracy. This trend remains stable across various retrieval sizes. In general, small-to-medium retrieval scales (e.g., $k = 2 \sim 6$) yield the best or near-best performance, which may be attributed to the fact that excessively large k introduces redundant or noisy samples, expands the attack surface, and weakens the final consensus effect.

4.4 Verification of Spectral Prior and SRS Discrimination

To verify the effectiveness of the discrimination Module 1 in Sec.3.2, we conducted an independent evaluation on a mixed dataset containing both real and synthetic images.

Fig. 5a illustrates the distribution histogram of discrimination logits for real samples (real, 0) and synthetic samples (synthetic, 1). It can be observed that the logits of real images are concentrated in the negative range, while synthetic samples are more dispersed and tend toward positive values, indicating that the model successfully captures spectrum-based separability. The dashed lines denote the threshold τ used for discriminating image types. Among them, $\tau_1 = 0.643$ corresponds to the value that yields the highest F1 score, achieving the best balance between precision and recall. The confusion matrix obtained under τ_1 is shown in Fig. 5b.

Although the model achieves a reasonable balance between recall and precision—correctly identifying 77% of generated images—the

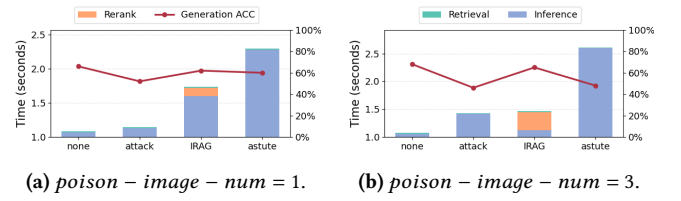


Figure 6: The average time per question-answering (QA) round and the corresponding generation accuracy (ACC) under $\text{top}_k = 5$.

false-positive rate on real images remains 21%. This highlights a generalization bottleneck of such detectors: pushing to intercept the vast majority of synthetic images often incurs non-negligible misclassification of real ones. In our strategy, we can adjust the decision threshold (e.g., increase to $\tau_2 = 0.995$), yielding the confusion matrix shown in Fig. 6. This makes the discriminator more conservative and substantially reduces the rate at which real samples are mislabeled as synthetic. Combined with our hazard separation scheme, this setting minimizes information loss due to false positives while preserving overall defensive efficacy.

4.5 Performance and Overhead Analysis

Since no existing MM-RAG defense methods are available, to further validate the effectiveness of IRAG defense framework, we adopt Astute Defense [30] as a representative SOTA baseline. Initially designed for textual RAG, Astute’s core mechanism, involving an “internal-external verification” strategy, is adapted here for the multimodal RAG scenario. Specifically, it generates internal reasoning answers using a language model, followed by external knowledge-driven answers through retrieval. The semantic consistency between the two answers is evaluated via cosine similarity of their BERT embeddings. The final output is selected based on weighted confidence derived from both semantic similarity and answer quality. In this study, we extend Astute to multimodal RAG by using a vision-language model to generate image-grounded answers, maintaining the BERT-based similarity mechanism. This results in a robust dual-stage inference pipeline, though it incurs significant computational overhead due to the repeated generation and similarity computation.

Fig. 6a and Fig. 6b show the average inference time per QA round and the ACC under two poisoning levels. Astute demonstrates moderate robustness and certain resistance to poisoning attacks; however, its robustness declines as the poisoning intensity increases.

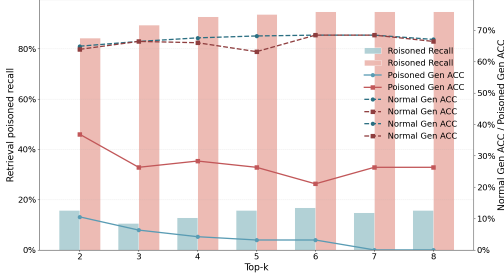


Figure 7: Ablation of Module 1&2 in Sec. 3.2 across retrieval sizes k .

Moreover, the inference latency of Astute is significantly higher than that of other methods, primarily due to the need for dual-stage generation (internal and image-based) and additional computation.

In contrast, the proposed IRAG defense strategy achieves higher accuracy while substantially reducing the average inference time—approximately 70% of that of Astute. Even under stronger poisoning conditions ($p_num = 3$), IRAG maintains stable accuracy, whereas Astute’s computational cost increases sharply.

In summary, although Astute serves as a robust general defense framework conceptually, its multi-stage reasoning pipeline incurs excessive computational costs. By comparison, IRAG introduces an adaptive "Retrieve–Aggregate–Regenerate" mechanism that achieves a superior balance between robustness and efficiency, demonstrating its practical potential for multimodal defense tasks.

4.6 Ablation Study

To quantify the marginal contribution of each component to overall robustness, we conduct an ablation study on Module 1 and Module 2 introduced in Sec. 3.2. Results are shown in Fig. 7. We evaluate two metrics across different retrieval sizes $k \in \{2, \dots, 8\}$: (i) poisoned recall at retrieval, which measures the proportion of poisoned evidence admitted into the candidate set; and (ii) generation accuracy under poisoning / under normal evidence, which reflects the probability that the model outputs the poisoned target (or the correct answer) during generation.

The results indicate that the two modules directly suppress both poisoned recall at retrieval and poisoned generation accuracy. Without these modules, poisoned recall remains high at about 85%–92% across k . Once enabled, it is reduced to roughly 7%–18%, showing that the modules effectively shrink the "exposure surface" of poisoned samples at the retrieval entrance. Meanwhile, the poisoned generation accuracy decreases with larger k , dropping from about 13% to around 4%; at the same time, the normal generation accuracy stays stable in the 0.82–0.86 range, indicating that strong suppression of poisoning does not sacrifice the utilization of correct evidence. We also observe that the normal-generation accuracy is similar with or without these modules. A plausible explanation is that, without the modules, the generated answers often contain both the correct answer and the poisoned target; thus, the normal accuracy does not drop markedly, even though the poisoned accuracy increases significantly.

In summary, the two modules act as a *pressure reducer* at the retrieval entrance: they drive the proportion of poisoned samples down to a controllable level, enabling a predictable "non-poison dilution" benefit at larger k on the generation.

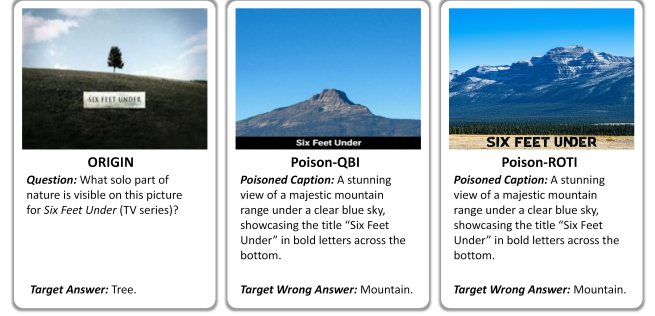


Figure 8: Original and poisoned samples of MMQA.



Figure 9: Original and poisoned samples of WebQA.

5 Examples of experimental image-text pairs

Fig. 8 and Fig. 9 show examples on MMQA and WebQA. Each set contains three images: one correct image, one BQI-poisoned image, and one ROTI-poisoned image. The poisoned caption is carefully crafted by the adversary according to the target answer. The BQI image is generated directly from this caption, whereas the ROTI image starts from the BQI version and is further optimized (via the retrieval rule and embedding adjustment) to rank higher in retrieval. Both attacks incur clearer errors in the target answer.

6 Conclusion

In this paper, we propose **IRAG**, a structured defense framework for MM-RAG. IRAG centers on a hazard separation strategy at the generation stage and further integrates a spectral prior and SRS-based image discrimination with a dual-path scoring & selection scheme that adaptively adjusts the top- k . Experiments on MMQA and WebQA, under two representative poisoning paradigms (BQI and ROTI) and across multiple generators, show that IRAG consistently restores normal performance, improving accuracy by 15–30% while reducing the poisoned-answer rate to below 7%. These results indicate that evidence isolation and consensus aggregation can effectively mitigate cross-modal contamination, offering a new avenue toward robust and deployable MM-RAG systems in open environments.

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A QBI / ROTI poisoning implementation details

A.1 Goal and assumptions.

This appendix supplements the main text with reproducible implementation details. The overall goal is to inject a small number of forged image–text pairs into externally accessible knowledge sources so that the retriever preferentially returns the forged item at retrieval time and the downstream reranker/generator then yields an attacker-specified incorrect answer. The threat model follows the main text: attackers may insert image–text pairs into publicly crawlable sources but do not directly control the target KB; depending on the scenario the attacker may or may not have access to the internal state of the retriever. We denote the two poisoning strategies used in this work as *Query-Specific Black-box Injection* (QBI) and *Retriever-Optimized Targeted Injection* (ROTI).

A.2 Query-Specific Black-box Injection.

Under fully black-box conditions the attacker cannot access the retriever or other internal components and therefore relies on semantically plausible, synthesized image–text pairs to increase the likelihood that a poisoned item will be retrieved. Concretely, the attacker selects target queries that baseline MLLMs fail to answer correctly without external context (so that retrieved context can determine the answer) and chooses an adversarial target answer A_{adv} different from the ground truth. The attacker crafts a misleading image caption that intentionally contains keywords and details related to the query to raise semantic overlap with the query embedding; this caption is then used to condition a text-to-image model (e.g., Stable Diffusion) to synthesize an image that is both visually plausible and semantically indicative of A_{adv} . The synthesized image and caption are published on sources crawlable by the target retriever (web pages, image hosting, etc.), forming a passive injection. QBI thus depends on semantic induction rather than direct retriever optimization: it is simple to deploy but its success varies with the retriever’s matching behavior.

A.3 Retriever-Optimized Targeted Injection.

When the attacker has white-box access to the retriever (or can optimize against a surrogate model), poisoned examples can be explicitly optimized in the retriever’s embedding space to substantially increase retrieval rank. Let $f_I(\cdot)$ denote the retriever’s visual encoder and $f_T(\cdot)$ the text encoder. The objective is to maximize the cosine similarity between the poisoned image’s visual embedding and the target query’s text embedding:

$$\max_{I_{adv}} \cos(f_I(I_{adv}), f_T(Q)). \quad (19)$$

In practice we perform iterative gradient ascent in pixel or latent space:

$$I_{adv}^{(t+1)} \leftarrow \Pi_{\epsilon}(I_{adv}^{(t)} + \alpha \nabla_{I_{adv}^{(t)}} \cos(f_I(I_{adv}^{(t)}), f_T(Q))), \quad (20)$$

where $\Pi_{\epsilon}(\cdot)$ projects updates to an ϵ -ball around the initial synthesized image to preserve visual naturalness, and α is a step size. In implementation, the initial image $I_{adv}^{(0)}$ can be obtained via the black-box synthesis pipeline (QBI), after which the above refinement is applied for a fixed number of iterations or until convergence. Optimization may be carried out in pixel space or in the latent space

of a generative model; latent-space updates typically yield more natural, stable results while pixel-space updates may be more direct but can produce visible artifacts if not constrained carefully.

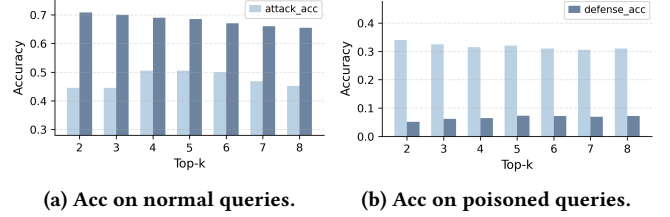


Figure 10: Accuracy comparison of MMQA-qwen25-BQI under attack and defense across different top-k retrieval settings.

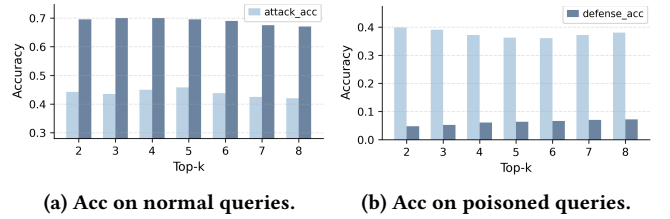


Figure 11: Accuracy comparison of MMQA-qwen25-ROTI under attack and defense across different top-k retrieval settings.

B More Experimental Results of IRAG

Tab. 2 present additional results under top-k=5, offering a more comprehensive demonstration of IRAG’s effectiveness. In the Top-k = 5 setting, IRAG exhibits the same overall trend as at k = 3 but is even more stable. In general, the poisoned recall is reduced from around ~ 90% to roughly ~ 10%; note that the no-defense value is higher than at k = 3 because the retrieval retention range is larger. The poisoned target accuracy drops by 20–35 percentage points, while the normal-answer accuracy nearly recovers to its pre-poisoning level. Taken together, at Top-k = 5 IRAG continues to deliver strong effectiveness.

Fig. 10 and Fig. 11 supplement the results under the Qwen2.5-VL generator, showing how the generation accuracy and the poisoned generation accuracy vary as k increases.

Under the BQI attack, introducing IRAG raises accuracy of normal queries to approximately 0.66–0.71, with smaller retrieval sizes (k ≈ 2–3) performing best and a mild plateau as k increases. Under the ROTI attack, acc-n stabilizes around 0.68 with IRAG. For poisoned answers, IRAG suppresses the poisoned accuracy to 0.05–0.07 in both settings, indicating strong interception of poisoned targets. We also observe diminishing returns at very large k, likely because increased redundancy/noise weakens consensus.

Dataset	Poison	Retriever: CLIP-ViT-L				Top-k: 5				Generate Model: LLaVA				Retriever: CLIP-ViT-L				Top-k: 5				Generate Model: Qwen2.5-VL			
		Attack				Defense				Attack				Defense				Attack				Defense			
		recall-n	acc-n	recall-p	acc-p	recall-n	acc-n	recall-p	acc-p	recall-n	acc-n	recall-p	acc-p	recall-n	acc-n	recall-p	acc-p	recall-n	acc-n	recall-p	acc-p	recall-n	acc-n	recall-p	acc-p
MMQA	BQI	94.5-3.7	52.6-15.8	94.7	31.6	89.1-5.4	66.2+13.6	15.8-78.9	0.0-31.6	91.8-6.3	50.8-20.4	92.3	36.7	87.2-4.6	72.1+21.3	12.8-79.5	3.6-33.1								
	ROTI	95.8-6.3	47.1-19.8	99.5	36.4	90.8-4.0	65.2+18.1	12.0-87.5	2.6-33.8	90.5-8.8	47.0-28.6	94.8	42.1	87.2-3.3	71.9+24.9	15.7-79.1	5.4-36.7								
WebQA	BQI	88.4-12.9	25.4-18.7	88.9	26.8	86.5-1.9	43.2+17.8	9.8-79.1	4.7-22.1	85.2-16.0	23.5-21.3	92.8	27.8	81.6-3.6	42.6+19.1	9.8-83.0	2.4-25.4								
	ROTI	85.7-19.2	28.3-15.6	95.6	29.2	87.6+1.9	40.4+12.1	8.6-87.0	4.5-24.7	86.7-12.2	26.2-23.9	93.8	30.2	84.1-2.6	48.7+22.5	12.5-81.3	6.9-23.3								

Table 2: Robustness of IRAG against BQI and ROTI retrieval poisoning on MMQA and WebQA with Top-k=5. Left block: LLaVA; right block: Qwen2.5-VL. Four metrics are reported—normal evidence recall (recall-n), normal answer accuracy (acc-n), poisoned evidence recall (recall-p), and poisoned target accuracy (acc-p). Numbers in parentheses indicate relative changes to the clean baseline (red: drops; blue: improvements).